

MS&E 228 (Winter 2026) — Lecture 14 Notes

What Can We Do About Unobserved Confounding?

Vasilis Syrgkanis
Stanford University

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Readings. *Applied Causal Inference Powered by ML and AI*, §12.1–12.2. Key papers: Cinelli & Hazlett (2020), Chernozhukov, Cinelli, Newey, Sharma, & Syrgkanis (2021), Bellemare, Bloem, & Wexler (2024).

Jupyter Notebook. Sensitivity Analysis with Sensemakr and Debiased ML (Darfur Example).

1 Introduction: Confronting Unobserved Confounding

This chapter marks a critical transition in our study of causal inference. In the preceding lectures, we developed the powerful framework of Directed Acyclic Graphs (DAGs) and Non-Parametric Structural Equation Models (NP-SEMs), which gave us a principled, visual language for encoding domain knowledge and checking whether conditional exogeneity holds. We introduced d-separation and Single World Intervention Graphs (SWIGs) as a formal calculus for determining valid adjustment sets. The central promise of that framework was to transform conditional exogeneity from an opaque assumption into a verifiable conclusion.

However, the graphical analysis also revealed a sobering reality. In our running hospital messaging example, we discovered that the set of observable patient data X was *not* sufficient to block all backdoor paths between the treatment (sending a message, D) and the outcome (patient glucose time-in-range, Y). The culprit was an unobserved confounder—the physician’s workload, U —which influences both the decision to send a message and the quality of that message, thereby affecting the outcome. The SWIG analysis showed an open backdoor path $D \leftarrow U \rightarrow M \rightarrow Y$, meaning that conditional exogeneity fails.

This leaves us at a crossroads. We have a well-defined causal question and a rich dataset, but our primary identification strategy—conditioning on observables—has been undermined. What should we do? This is not an unusual situation; in fact, it is the norm in most observational studies. Unobserved confounders are present everywhere, and the honest analyst must confront them head-on.

The goals of this lecture are fourfold:

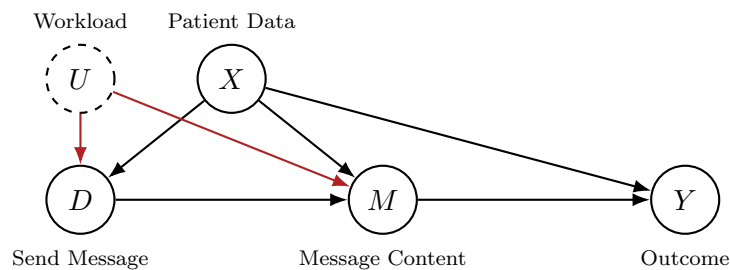
1. **Sensitivity Analysis:** Develop a framework to *bound* the true treatment effect by making explicit, interpretable assumptions about the strength of unobserved confounding.

2. **Practical Tools:** Learn how to make sensitivity analysis actionable through benchmarking against observed covariates, contour plots, and the robustness value.
3. **Move beyond conditional exogeneity:** Realize that conditional exogeneity is only one type of identification argument. There is no reason to restrict ourselves to one type of identification argument. The identification problem asks a much broader question: “can we the causal estimand as a function of observable quantities”.
4. **Alternative Identification:** Introduce the front-door criterion as an example of an identification strategy that does not rely on observing all confounders, broadening our toolkit beyond simple adjustment.

2 The Unobserved Confounder Problem: Where We Stand

Let us briefly recap the situation we arrived at in the previous lecture. We introduced the powerful framework of Directed Acyclic Graphs (DAGs) and non-parametric structural equation models. We used the visual tool of Single World Intervention Graphs (SWIGs) to identify what are good controls and what are bad controls, allowing us to check when conditional exogeneity holds. The conditional exogeneity assumption, which we have been using throughout this course, posits that there exists a set of variables that we observe and can control for such that the treatment is as good as random, conditional on those variables.

In the clinical example, this assumption failed when we were interested in estimating the effect of sending a message (D) on patient outcomes (Y). The causal graph for this setting, which we developed in the previous lecture, is shown below.



The red arrows highlight the confounding problem: the physician’s workload U affects both the treatment decision D and the message content M , creating an open backdoor path that cannot be blocked by conditioning on the patient data X alone.

What Would You Do Next?

“You’ve talked to the group of physicians and collected the data from the monitoring system. You go back to the drawing board with your causal inference colleagues, and you realize there’s this problem. What do you do next?” This is a fundamental question that every applied researcher must face. Before reading on, consider what options you would explore.

2.1 Our Options When Exogeneity Fails

When the exogeneity assumption fails due to unobserved confounding, we have several paths forward:

1. **Go back and collect the missing data.** The first and most natural response is to ask whether the unobserved confounder can, in fact, be measured. In our hospital example, this means asking: are there variables describing the physician’s activities that were simply forgotten? Perhaps there is a physician scheduling table or a workload log that was not incorporated into the dataset. This is often the first thing to consider. However, the practical answer is frequently either “no, that data does not exist” or “it would take years to get the necessary approvals and data infrastructure in place.” While this avenue should always be explored, we need strategies for the common case where it is not immediately feasible.
2. **Conduct a sensitivity analysis.** Rather than giving up, we can try to *quantify* the potential impact of the unobserved confounder. We must be careful with the term “quantify” here, because we cannot truly estimate something we do not observe or measure from the data. Instead, we can *posit* the potential effect of the confounder based on domain knowledge, quantifying how much we believe the unobserved confounder might explain the outcome or the treatment. This allows us to produce **bounds** on the true treatment effect, rather than a single point estimate. The width of these bounds reflects our uncertainty about the strength of the confounding.
3. **Find a new identification strategy.** Perhaps the most intellectually exciting option is to ask whether there is a more clever way to identify the treatment effect that does not require observing all confounders. As we will see later in this lecture with the front-door criterion, and in subsequent lectures with instrumental variables, there are indeed powerful identification strategies that exploit specific features of the causal graph to bypass unobserved confounding entirely.

These three approaches—collecting more data, conducting a sensitivity analysis, or finding an alternative identification strategy—are the main paths forward. For the bulk of this lecture, we will focus on the second option, sensitivity analysis, and will conclude with an introduction to the front-door criterion as an example of the third option. We will explore other identification strategies, particularly instrumental variables, in the next lecture.

3 Part I: Sensitivity Analysis — The Simple Case

We begin our study of sensitivity analysis with the simplest possible setting. By making some strong but clarifying assumptions, we will derive the fundamental structure of the omitted variable bias and build intuition that will carry over to the general case.

3.1 A Partially Linear Model with a Confounder

Consider a setting with a treatment D , an outcome Y , observed covariates X , and a critical unobserved confounder U . To make this concrete in our running clinical example: D is the decision to send a message to a patient, Y is the patient’s glucose time-in-range, X represents all the data collected from the monitoring system, and U is the physician’s workload.

For the sake of simplifying the derivations, we will assume a **partially linear model**. Specifically, while the observed covariates X can affect both the treatment D and outcome Y in arbitrary, non-linear ways, the relationships involving the unobserved confounder U and the treatment D are assumed to be linear. This gives us the following structural equation model:

Partially Linear Model with Unobserved Confounding

$$Y = \theta D + \delta U + g(X) + \varepsilon_Y, \quad \mathbb{E}[\varepsilon_Y \mid D, X, U] = 0 \quad (1)$$

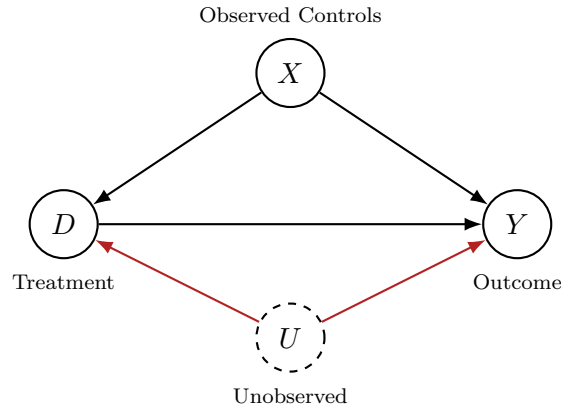
$$D = \gamma U + f(X) + \varepsilon_D, \quad \mathbb{E}[\varepsilon_D \mid X, U] = 0 \quad (2)$$

where $\text{Var}(U) = 1$ and $\mathbb{E}[U] = 0$ (normalization), and $g(\cdot)$ and $f(\cdot)$ are arbitrary unknown functions.

Several aspects of this model deserve comment. The functions $g(X)$ and $f(X)$ capture the potentially complex, non-linear effects of the observed covariates on the outcome and treatment, respectively. The key simplification is that the unobserved confounder U enters both equations linearly, with coefficients δ (its direct effect on the outcome) and γ (its direct effect on the treatment).

The normalization $\text{Var}(U) = 1$ is a standard assumption for model identifiability. If the variance of U were some other value σ_U^2 , we could always redefine $\delta' = \delta \sigma_U$ and $\gamma' = \gamma \sigma_U$ and work with the standardized confounder $U' = U/\sigma_U$. The parameters δ and γ would simply absorb the scale. So we proceed with the understanding that our unobserved confounder is standardized.

The corresponding DAG for this simplified model is:



Our primary objective is to estimate the true causal effect θ . To correctly identify this effect,

we would need to condition our analysis on both the observed covariates X and the unobserved confounder U . This leads to the crucial question: what happens if we simply omit U from our model and proceed with the tools we have developed so far?

3.2 What Does Double ML Estimate When U Is Omitted?

Suppose we do not have access to U and we condition only on the observed covariates X . Because we have a partially linear structural equation model, the natural algorithm to use is the Double Machine Learning (DML) algorithm from earlier in the course. We have already established what DML estimates: it converges to the population OLS coefficient that relates the “surprises” in the outcome to the “surprises” in the treatment, after we have removed everything that is predictable from the observed controls X .

Formally, DML estimates the coefficient θ_s in the population regression:

$$\theta_s = \frac{\mathbb{E}[\tilde{Y} \cdot \tilde{D}]}{\mathbb{E}[\tilde{D}^2]}$$

where $\tilde{Y} = Y - \mathbb{E}[Y | X]$ is the outcome residual and $\tilde{D} = D - \mathbb{E}[D | X]$ is the treatment residual. The question is: how does this estimate θ_s relate to the true treatment effect θ ?

To answer this, we can simplify our equations by **residualizing** them with respect to X . As we saw in the Frisch–Waugh–Lovell (FWL) lecture, residualization is a linear operation. Residualizing the outcome equation (1) with respect to X corresponds to residualizing every single element on the right-hand side with respect to X . Since $g(X)$ is a function of X alone, its residual is zero, and X effectively drops out of the equations. We are left with a much simpler linear system:

Residualized System

After residualizing with respect to X :

$$\tilde{Y} = \theta \tilde{D} + \delta U + \varepsilon_Y \quad (3)$$

$$\tilde{D} = \gamma U + \varepsilon_D \quad (4)$$

where $\tilde{Y} = Y - \mathbb{E}[Y | X]$, $\tilde{D} = D - \mathbb{E}[D | X]$, and we have used the fact that U is independent of X in this simplified model (or more precisely, $\tilde{U} = U - \mathbb{E}[U | X]$ replaces U , but for notational simplicity we keep U).

This residualized system is the key to understanding the omitted variable bias. The DML algorithm is estimating the OLS coefficient of $\tilde{Y}$ on $\tilde{D}$ in this system. Is that coefficient θ ? No, it should not be, because U and $\tilde{D}$ are correlated through the treatment equation (4). The OLS coefficient will pick up not only the true causal effect θ but also an extra term arising from this correlation.

3.3 Deriving the Omitted Variable Bias

Let us now carefully derive the bias. We need to compute the population OLS coefficient:

$$\theta_s = \frac{\mathbb{E}[\tilde{Y} \cdot \tilde{D}]}{\mathbb{E}[\tilde{D}^2]}$$

Step 1: Expand the numerator. Substituting the residualized outcome equation (3):

$$\begin{aligned}\mathbb{E}[\tilde{Y} \cdot \tilde{D}] &= \mathbb{E}[(\theta \tilde{D} + \delta U + \varepsilon_Y) \cdot \tilde{D}] \\ &= \theta \mathbb{E}[\tilde{D}^2] + \delta \mathbb{E}[U \cdot \tilde{D}] + \mathbb{E}[\varepsilon_Y \cdot \tilde{D}]\end{aligned}\tag{5}$$

Step 2: Simplify the noise term. The exogenous noise ε_Y is independent of the treatment D (and hence of $\tilde{D}$) by the structural equation model. Therefore:

$$\mathbb{E}[\varepsilon_Y \cdot \tilde{D}] = 0$$

Step 3: Compute $\mathbb{E}[U \cdot \tilde{D}]$. This is the crucial term that generates the bias. Using the treatment equation (4):

$$\begin{aligned}\mathbb{E}[U \cdot \tilde{D}] &= \mathbb{E}[U \cdot (\gamma U + \varepsilon_D)] \\ &= \gamma \mathbb{E}[U^2] + \mathbb{E}[U \cdot \varepsilon_D] \\ &= \gamma \cdot 1 + 0 = \gamma\end{aligned}\tag{6}$$

where we used the normalization $\text{Var}(U) = \mathbb{E}[U^2] = 1$ (and $\mathbb{E}[U] = 0$) and the independence of U and ε_D .

Step 4: Assemble the bias formula. Substituting equations (6) and the zero noise term back into (5), and dividing by $\mathbb{E}[\tilde{D}^2]$:

$$\theta_s = \frac{\theta \mathbb{E}[\tilde{D}^2] + \delta \gamma}{\mathbb{E}[\tilde{D}^2]} = \theta + \frac{\delta \gamma}{\mathbb{E}[\tilde{D}^2]}$$

Omitted Variable Bias Formula (Simple Case)

The DML estimate converges to:

$$\theta_s = \theta + \underbrace{\frac{\delta\gamma}{\mathbb{E}[\tilde{D}^2]}}_{\text{Omitted Variable Bias}}$$

where:

- δ = direct effect of U on Y (how much the confounder affects the outcome),
- γ = direct effect of U on D (how much the confounder affects the treatment),
- $\mathbb{E}[\tilde{D}^2]$ = residual variance of the treatment after partialling out X .

3.4 Interpreting the Bias Formula

The omitted variable bias formula has a beautifully intuitive structure. The bias is the product of three quantities:

1. **The confounder's effect on the outcome (δ):** How much does the unobserved confounder directly influence the outcome, on top of the treatment and observed controls? In our hospital example, this asks: how much does the physician's workload affect the patient's outcome (through improving the efficacy of the message)?
2. **The confounder's effect on the treatment (γ):** How much does the unobserved confounder influence the treatment assignment, on top of the observed controls? In our example: how much does the physician's workload affect whether they decide to send a message?
3. **The residual treatment variation ($\mathbb{E}[\tilde{D}^2]$):** This is the total variation in the treatment that remains after removing everything predictable from X . This quantity is estimable from the data.

Key Insight: The Bias is a Product

The bias depends on the **product** $\delta \cdot \gamma$. This means that even if we have an unobserved confounder that is very predictive of the outcome (large $|\delta|$), if it has almost nothing to do with the treatment (small $|\gamma|$), the resulting bias will be small. Conversely, a confounder that strongly predicts the treatment but has little direct effect on the outcome will also produce small bias. To create a large bias, the unobserved confounder must be strongly predictive of **both** the treatment and the outcome, even on top of all the variables we have already controlled for.

3.5 From Point Estimates to Bounds

How can we use this bias formula in practice? If we knew the exact values of δ and γ , we could directly calculate and subtract the bias. However, it is unrealistic to assume we know these parameters precisely—after all, they involve an unobserved variable. A more believable scenario is that we can specify a *range* for them by consulting domain experts.

Consulting Domain Experts

In the hospital setting, we could ask the physicians:

- “How large do you believe the direct influence of a physician’s workload is on their treatment decision?” They might respond that this influence is minimal because the decision is primarily driven by patient monitoring data. They might quantify this by saying that the workload won’t increase the probability of treatment by more than 10%, so we can set a bound $|\gamma| \leq 0.1$.
- “How much does the physician’s workload affect the quality of the message, and does this variation significantly change the message’s efficacy?” The expert might believe this effect is also small, perhaps changing the outcome by at most 5%. This gives us $|\delta| \leq 0.05$.

The denominator $\mathbb{E}[\tilde{D}^2]$ represents the variation of the treatment “surprise”—the variation left after accounting for all the control variables—which is something we can estimate from the data. Given these components, we can establish that the true treatment effect lies within an interval:

$$\theta \in \left[\hat{\theta}_s - \frac{|\delta|_{\max} \cdot |\gamma|_{\max}}{\mathbb{E}_n[\tilde{D}^2]}, \quad \hat{\theta}_s + \frac{|\delta|_{\max} \cdot |\gamma|_{\max}}{\mathbb{E}_n[\tilde{D}^2]} \right]$$

This is no longer a point estimate but an **interval**. Importantly, even with infinite data, we would still have an interval. This is not a typical confidence interval that shrinks to a point as the sample size grows. Instead, it is a **partial identification interval**, reflecting the fundamental lack of information about the unobserved confounder.

This was the simplest version of the sensitivity analysis framework. One might argue that the linearity assumptions we have been making are too strong. A natural question is: does this style of analysis extend to more general structural equation models? The answer, as we will see next, is yes.

4 Part II: General Sensitivity Analysis — From Linear Coefficients to Explained Variance

4.1 Generalizing Beyond Linearity

So far, we have been making strong linearity assumptions about how the unobserved confounder enters the structural equations. A natural question is whether this style of analysis extends to more general structural equation models. Let us explore what we can do within the framework of general, non-parametric structural equations.

We now assume a much more permissive model. We have an unobserved confounder U , which can be multivariate, and a general structural equation framework:

General Partially Linear Model with Unobserved Confounding

Consider the following partially linear non-parametric structural equation model (noise terms are mutually independent):

$$Y(d, x, u) = \theta \cdot d + f_Y(x, u, \varepsilon_Y) \quad (7)$$

$$D(x, u) = f_D(x, u, \varepsilon_D) \quad (8)$$

which generates observed variables, based on:

$$Y = \theta \cdot D + f_Y(X, U, \varepsilon_Y) \quad (9)$$

$$D = f_D(X, U, \varepsilon_D) \quad (10)$$

The key assumptions implied by this structural equation model are:

1. The potential outcome $Y(d)$ is independent of the treatment D given X and U :

$$Y(d) \perp\!\!\!\perp D \mid X, U.$$

2. The conditional expectation of Y given D , X , and U is partially linear in D :

$$\mathbb{E}[Y \mid D, X, U] = \theta D + g(X, U)$$

This model is much more permissive than the previous one. The confounder U can be multivariate and can interact with X and the idiosyncratic noise terms in arbitrary ways through the functions $f_Y(X, U, \varepsilon_Y)$ and $f_D(X, U, \varepsilon_D)$. The only structural restriction is that the treatment effect θ is constant (partial linearity in D). We will relax even this assumption later when we discuss binary treatments.

Now suppose that instead of running the full specification (which would require observing U), we run Double ML while omitting U and converge to some quantity θ_s . Can we still characterize the

resulting omitted variable bias? Or was the simple formula we derived earlier just a happenstance of the linear framework?

4.2 Measuring Confounding Strength via Explained Variance

In the simple linear case, the bias was characterized by the coefficients δ and γ . In the general case, a natural question arises: are there analogues to these coefficients that are still interpretable? Can we define similar quantities such that we can ask a domain expert to provide bounds on them, and in turn characterize the omitted variable bias?

The answer is yes. We can still establish a product formula to characterize the bias. However, instead of the simple coefficients δ and γ , we need to employ a more general measure of predictive power. Specifically, we need a measure that quantifies the predictive power of the unobserved confounder on both the outcome and the treatment, conditioned on all the observed variables. This measure is captured by the concept of the **partial R^2** , which precisely measures the relative gain in explained variance that we obtain by adding the unobserved confounder to our predictive model.

4.2.1 Strength of the Confounder on the Outcome: η_Y^2

Consider the outcome Y . If we had the best possible predictive model for Y given the treatment D , the observed confounders X , and the unobserved confounder U , we would still have some residual, unexplained variation. This remnant variation, $\mathbb{E}[\text{Var}(Y \mid D, X, U)]$, cannot be explained by any of our variables. Now, if we only use D and X to predict the outcome (omitting U), we are explaining only a subset of the variation, leaving a larger amount of unexplained variation $\mathbb{E}[\text{Var}(Y \mid D, X)]$. The question we are asking is: what is the relative gain in explained variance from adding U ?

Definition: Partial R^2 of the Confounder on the Outcome

$$\eta_{Y \sim U \mid D, X}^2 = \frac{\mathbb{E}[\text{Var}(Y \mid D, X)] - \mathbb{E}[\text{Var}(Y \mid D, X, U)]}{\mathbb{E}[\text{Var}(Y \mid D, X)]} \in [0, 1]$$

Interpretation: The fraction of the residual outcome variance (after accounting for D and X) that is explained by the unobserved confounder U .

This is closely related to the standard R^2 measure used in model selection, but with a crucial difference. A standard R^2 compares the variance explained by the predictors to a model with no predictors (a constant model), which is why the overall variance appears in the denominator. Here, we are asking about the *extra* variance explained by a set of predictors (D, X, U) compared to a subset of those predictors (D, X). This is why we call it a **partial R^2** .

There are three equivalent ways to interpret this quantity:

1. **Gain in explained variance:** How much more of the outcome's variation can we explain by adding U to a model that already includes D and X ?

2. **Reduction in residual variance:** By what fraction does the residual variance of Y decrease when we add U to the model?
3. **Predictive power of U :** How much extra predictive power does U have for the outcome, beyond what D and X already provide?

4.2.2 Strength of the Confounder on the Treatment: η_D^2

Similarly, we can define the predictive power of U for the treatment. This measures how much extra variance of the treatment is explained by the unobserved confounder U compared to a model that only uses the observed confounders X .

Definition: Partial R^2 of the Confounder on the Treatment

$$\eta_{D \sim U|X}^2 = \frac{\mathbb{E}[\text{Var}(D | X)] - \mathbb{E}[\text{Var}(D | X, U)]}{\mathbb{E}[\text{Var}(D | X)]} \in [0, 1]$$

Interpretation: The fraction of the residual treatment variance (after accounting for X) that is explained by the unobserved confounder U .

Note the subtle difference in the conditioning sets: for the outcome, we condition on both D and X (because we want to know the extra predictive power of U on top of both the treatment and the controls), while for the treatment, we condition only on X (because we want to know how much U helps predict the treatment on top of the observed controls).

These are the two quantities on which we need to provide bounds. They cannot be measured from the data because we do not observe U . However, we might be able to say, based on domain knowledge, that we believe the unobserved confounder can explain at most, say, 2% more of the variance of the outcome and at most 5% more of the variance of the treatment. These are our sensitivity parameters.

4.3 The General Omitted Variable Bias Formula

With these definitions in hand, we can state the general bias characterization. The DML estimate θ_s , computed using only the observed controls X , satisfies:

General Omitted Variable Bias Formula

$$|\theta_s - \theta|^2 \leq \eta_{Y \sim U|D,X}^2 \cdot \frac{\eta_{D \sim U|X}^2}{1 - \eta_{D \sim U|X}^2} \cdot \underbrace{\frac{\mathbb{E}[\text{Var}(Y | D, X)]}{\mathbb{E}[\text{Var}(D | X)]}}_{\text{Estimable from data: } S^2}$$

Let us unpack this formula. The squared bias is bounded by a product of three terms:

1. $\eta_{Y \sim U|D,X}^2$: The partial R^2 of the confounder on the outcome. This is the qualitative analogue of δ^2 from the simple case.

2. $\eta_{D \sim U|X}^2$: A monotone transformation of the partial R^2 of the confounder on the treatment. This is the qualitative analogue of γ^2 from the simple case.
3. $S^2 = \frac{\mathbb{E}[\text{Var}(Y|D,X)]}{\mathbb{E}[\text{Var}(D|X)]}$: A scaling factor that can be estimated from the data. The numerator is the unexplained variance of the outcome Y after subtracting everything predictable from the treatment D and the controls X —this is simply the variance of the final OLS residual in the DML algorithm. The denominator is the unexplained variance of the treatment D after subtracting everything predictable from the controls X —this is the variance of the residualized treatment. This is the qualitative analogue of $1/\mathbb{E}[\tilde{D}^2]$ from the simple case.

Since the scaling factor S^2 can be estimated from the data, the two quantities on which we need to posit bounds are η_Y^2 and η_D^2 . This is how we operationalize the sensitivity analysis.

Footnote: Pattern Matching the Simple and General Formulas

To see how the general formula reduces to the simple case, recall that in the simple linear model, $\mathbb{E}[\text{Var}(Y | D, X)] = \delta^2 \text{Var}(U) + \text{Var}(\varepsilon_Y) = \delta^2 + \sigma_Y^2$ and $\mathbb{E}[\text{Var}(Y | D, X, U)] = \text{Var}(\varepsilon_Y) = \sigma_Y^2$. Therefore:

$$\eta_Y^2 = \frac{\delta^2 + \sigma_Y^2 - \sigma_Y^2}{\delta^2 + \sigma_Y^2} = \frac{\delta^2}{\delta^2 + \sigma_Y^2}$$

Similarly, $\eta_D^2 = \gamma^2/(\gamma^2 + \sigma_D^2)$, and the scaling factor becomes $S^2 = (\delta^2 + \sigma_Y^2)/(\gamma^2 + \sigma_D^2)$. Substituting these into the general formula and simplifying recovers the simple bias formula $|\theta_s - \theta|^2 = |\delta\gamma|^2/\mathbb{E}[\tilde{D}^2]^2 = |\delta\gamma|/(\gamma^2 + \sigma_D^2)^2$.

5 Part III: The Sensitivity Analysis Procedure — From Theory to Practice

We now have all the theoretical ingredients to conduct a sensitivity analysis. In this section, we describe the step-by-step procedure for turning the bias formula into actionable bounds on the treatment effect.

5.1 The Four-Step Procedure

The sensitivity analysis procedure consists of four steps:

The Sensitivity Analysis Procedure

Step 1: Run DML on the observed controls. Run the Double Machine Learning algorithm using only the observed control variables X . This produces:

- A (biased) point estimate $\hat{\theta}_s$.
- The estimated outcome residuals $\check{Y}_i = Y_i - \hat{g}(X_i)$ and treatment residuals $\check{D}_i = D_i - \hat{f}(X_i)$.

Step 2: Estimate the scaling factor S^2 . From the residuals, compute:

- The variance of the treatment residual: $\hat{\nu}^2 = \frac{1}{n} \sum_{i=1}^n \check{D}_i^2$.
- The variance of the final OLS residual: $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\check{Y}_i - \hat{\theta}_s \check{D}_i)^2$.
- The scaling factor: $\hat{S}^2 = \hat{\sigma}^2 / \hat{\nu}^2$.

Step 3: Elicit bounds on the sensitivity parameters. Consult domain experts or use benchmarking (described in the next section) to obtain upper bounds $\bar{\eta}_Y^2$ and $\bar{\eta}_D^2$ on the partial R^2 values of the unobserved confounder.

Step 4: Compute the sensitivity bounds. The bias is bounded by:

$$|\text{bias}| \leq \hat{S} \cdot \sqrt{\bar{\eta}_Y^2 \cdot \frac{\bar{\eta}_D^2}{1 - \bar{\eta}_D^2}}$$

The true treatment effect θ lies in the interval:

$$\theta \in \left[\hat{\theta}_s - |\text{bias}|, \quad \hat{\theta}_s + |\text{bias}| \right]$$

Two important questions arise from this procedure. First, at most how much more of the variance of the outcome could you explain with unobserved confounders? This gives us the upper bound $\bar{\eta}_Y^2$. Second, at most how much more of the variance of the treatment could you explain? This gives us $\bar{\eta}_D^2$. We can then plug these values into the bias formula to obtain the partial identification interval.

It is important to understand that even with infinite data, we would still have an interval. This is not a typical confidence interval that would shrink to a point with more data. Instead, it is a **partial identification interval**. This interval exists because we are lacking information—specifically, we are lacking variables. The interval reflects fundamental uncertainty about the unobserved confounder, not statistical uncertainty about finite samples.

5.2 Confidence Bounds

In practice, we must also account for the statistical uncertainty in our estimates. Both the DML estimate $\hat{\theta}_s$ and the scaling factor $\hat{S}$ are estimated from finite data and therefore have sampling variability. To construct bounds that hold with, say, 95% confidence, we need to incorporate standard errors.

Confidence Bounds for θ

A 95% confidence interval for the true treatment effect is:

$$\left[\hat{\theta}_{\text{lower}} - 1.96 \cdot \text{SE}_{\text{lower}}, \quad \hat{\theta}_{\text{upper}} + 1.96 \cdot \text{SE}_{\text{upper}} \right]$$

where $\hat{\theta}_{\text{lower}}$ and $\hat{\theta}_{\text{upper}}$ are the bounds from the sensitivity analysis, and the standard errors account for the uncertainty in estimating both $\hat{\theta}_s$ and $\hat{S}$.

While the exact formula for the standard errors is somewhat involved, the underlying intuition is the same as what we used in the DML insensitivity theory. Because DML has the insensitivity property (i.e., it is robust to small perturbations in the nuisance function estimates), we can produce confidence intervals for both the lower and upper bounds using simple plug-in formulas. The standard errors for the DML component are exactly what we derived in earlier lectures, and the additional uncertainty from the scaling factor can be incorporated using similar techniques. The exact formulas for the standard errors are derived using the delta method, as detailed in Appendix A.

5.3 Pseudocode Implementation

Here is a rough outline of how to implement the sensitivity bounds in Python:

Python: Computing Sensitivity Bounds

```
import numpy as np

def dml_sensitivity_bounds(res, eta_ysq, eta_dsqr):
    """
    Compute sensitivity bounds for the treatment effect.

    Parameters:
        res: tuple of (y_residuals, D_residuals) from DML
        eta_ysq: upper bound on partial  $R^2$  of U on Y
        eta_dsqr: upper bound on partial  $R^2$  of U on D

    Returns:
        (lower_bound, upper_bound) for the treatment effect
    """
    y_res, D_res = res

    # Step 1: Estimate treatment residual variance
    nu_sq = np.mean(D_res ** 2)
    # Step 2: Compute biased point estimate
    theta_s = np.mean(y_res * D_res) / nu_sq
    # Step 3: Compute outcome residual variance
    sigma_sq = np.mean((y_res - D_res * theta_s) ** 2)
    # Step 4: Compute scaling factor
    S = np.sqrt(sigma_sq / nu_sq)
    # Step 5: Compute confounding factors
    C_d_sq = eta_dsqr / (1 - eta_dsqr)
    C_g_sq = eta_ysq
    # Step 6: Compute bias bound
    error = S * np.sqrt(C_d_sq * C_g_sq)

    # Step 7: Calculate standard errors for lower and upper bound
    psi_theta = (y_res - D_res * theta_s) * D_res / nu_sq # influence function for theta_s
    psi_sigmasq = (y_res - D_res * theta_s)**2 - sigma_sq # influence function for sigma_sq
    psi_nusq = D_res**2 - nu_sq # influence function for nu_sq

    # delta method to calculate the influence function of the lower and upper bounds
    # Roughly: sum of derivatives of lower / upper bound formula wrt
    # to each parameter times the corresponding influence function
    phi_plus = psi_theta
    phi_plus += (np.sqrt(C_d_sq * C_g_sq) / (2 * S)) * (-(sigma_sq / (nu_sq**2)) * psi_nusq
        + (1 / nu_sq) * psi_sigmasq)
    stderr_plus = np.sqrt(np.mean(phi_plus**2) / phi_plus.shape[0])

    phi_minus = psi_theta
    phi_minus -= (np.sqrt(C_d_sq * C_g_sq) / (2 * S)) * (-(sigma_sq / (nu_sq**2)) *
        psi_nusq + (1 / nu_sq) * psi_sigmasq)
    stderr_minus = np.sqrt(np.mean(phi_minus**2) / phi_minus.shape[0])

    return theta_s - error, theta_s + error
```

A full implementation of this methodology, applied to the Darfur conflict data, is available in the linked Jupyter notebook. In that application, the goal is to understand the effect of prior direct exposure to war on a person’s attitude toward peace. Most of the available datasets for this problem only have demographic information (sex, age, household size, etc.) as controls. Obviously, there could be many other unobserved confounders that correlate a person’s prior exposure to war and their attitude towards peace, making sensitivity analysis essential.

6 Part IV: Making Sensitivity Practical — Benchmarking and the Robustness Value

A key challenge in operationalizing the sensitivity analysis framework is coming up with plausible bounds for the partial R^2 values η_Y^2 and η_D^2 . How can we come up with reasonable, ballpark estimates for the predictive power of an unobserved confounder when, by definition, we have never observed it? In this section, we describe several practical strategies.

6.1 Benchmarking Against Observed Covariates

The most practical and widely used approach is to **benchmark** the unobserved confounder against the observed covariates. The idea is simple: for each observed control variable X_j (or group of controls), we can calculate how much explained variance it adds to the outcome and treatment on top of the other controls. This gives us a concrete sense of scale for the η^2 values.

Benchmarking Procedure

1. For each observed covariate X_j (or group of covariates), compute its partial R^2 for the outcome:

$$\hat{\eta}_{Y \sim X_j | D, X_{-j}}^2 = \frac{\hat{V}(Y | D, X_{-j}) - \hat{V}(Y | D, X)}{\hat{V}(Y | D, X_{-j})}$$

2. Similarly, compute its partial R^2 for the treatment:

$$\hat{\eta}_{D \sim X_j | X_{-j}}^2 = \frac{\hat{V}(D | X_{-j}) - \hat{V}(D | X)}{\hat{V}(D | X_{-j})}$$

3. Use these as reference points. Ask: “If our unobserved confounder is as strong as variable X_j , would our conclusion still hold? What if it is twice as strong?”

For instance, in the Darfur example, we can take a set of observed controls like demographics (sex, age, farmer, herder, past-voted, household size) and calculate what percentage of the variance in the treatment (exposure to war) and the outcome (attitudes towards peace) they explain. If demographics explain 1% of the variance in the treatment and 7% of the variance in the outcome, we can conservatively assume that the unobserved confounder is as predictive as all of these de-

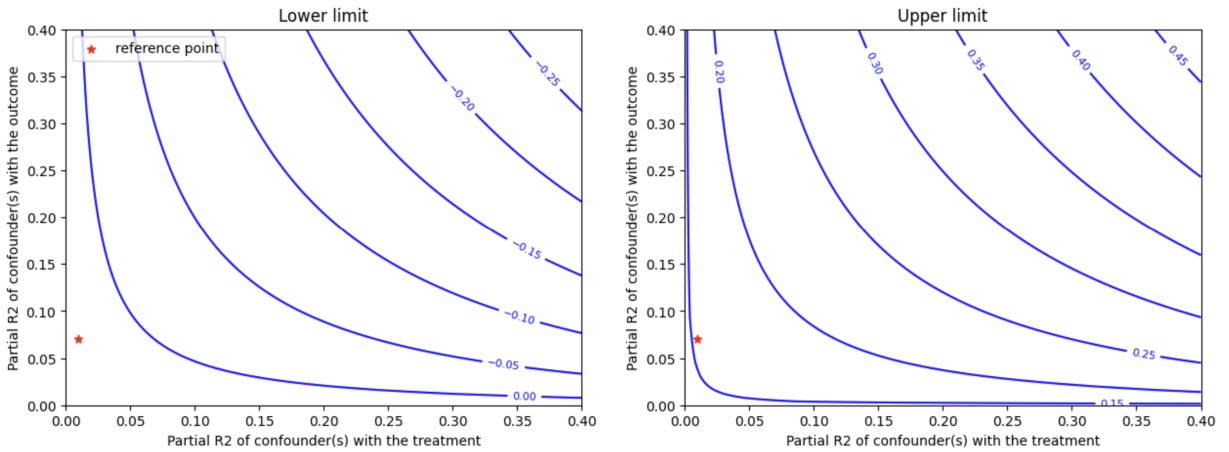
mographics combined. This gives us concrete values for η_Y^2 and η_D^2 to plug into our sensitivity analysis.

6.2 Contour Plots

Rather than providing a single sensitivity bound, a more informative approach is to produce a **contour plot**. On the x -axis, we plot the predictive power of the unobserved confounder on the treatment (its partial η_D^2 on top of the observed covariates X), and on the y -axis, we plot its predictive power on the outcome (its partial η_Y^2 on top of the treatment D and covariates X). For each combination of these two values, we can compute the resulting bound on the treatment effect.

In the Darfur analysis, controlling only for demographics yields a treatment effect of approximately 0.10 from a DML model. This suggests that prior exposure to war increased the probability of having a positive attitude towards peace by 10 percentage points. However, we must consider the possibility of unobserved confounders. Since we found a positive treatment effect, our main concern is whether an unobserved confounder could overturn this result by pushing the true effect below zero. We are therefore interested in the lower bound of the estimated effect, which is our estimated $\hat{\theta}_s$ minus the potential bias term. If this lower bound crosses zero, it becomes plausible that, had we observed the confounder, we would not have been able to claim a positive effect.

We can plot the contour of this lower bound—often the 95% confidence lower bound, which is $\hat{\theta}_s - \text{bias} - 1.96 \times \text{SE}$ —for all possible values of the confounder’s predictive powers. Since the bias is increasing in both η_Y^2 and η_D^2 , the contour line where the lower bound equals zero will be a curve in the (η_D^2, η_Y^2) plane. Any combination of confounding strengths below this curve leaves our qualitative conclusion intact; any combination above it would overturn our finding.



Furthermore, we can plot our benchmark points on this same graph. For example, we can calculate the predictive power of all demographic variables on the treatment (e.g., 1%) and on the outcome (e.g., 7%) and place that point on the plot. If this benchmark point is far away from the contour line that overturns the result, we can argue that our unobserved confounder would need to be substantially more predictive than all observed demographics to invalidate our conclusion.

This visualization allows the researcher and the decision-maker to reason about the strength of confounding required to change the result, using the observed covariates as a tangible yardstick.

6.3 The Robustness Value

For a more concise summary of the sensitivity analysis, we can report a single metric known as the **robustness value**.

Definition: Robustness Value (RV)

The robustness value is the value c^2 such that if the unobserved confounder explained c^2 of the residual variance in both the treatment and outcome (i.e., $\eta_Y^2 = \eta_D^2 = c^2$), the 95% confidence interval for the treatment effect would just include zero.

Geometrically, the robustness value corresponds to the point on the 45-degree line of the contour plot where the confidence bound crosses zero. It answers the question: assuming an unobserved confounder explains the same additional percentage of variance in both the treatment and the outcome, how large would that percentage need to be to invalidate our causal conclusion?

Interpreting the Robustness Value

If your robustness value is 5%, you can report: “An unobserved confounder would need to explain at least 5% of the residual variance in both treatment and outcome to change our conclusion that the effect is non-zero.” This provides a clear and intuitive measure of the finding’s robustness. You can then compare this to the benchmark values from observed covariates. If all observed demographics together explain less than 5% of the treatment variance, and you believe no single unobserved confounder could be more predictive than all demographics combined, your finding is robust.

It is important to remember that this entire analysis operates under the assumption of a partially linear model, which implies a constant treatment effect. In many applications, particularly those involving binary treatments, this assumption is too restrictive. We address this limitation next.

7 Part V: Beyond Partial Linearity — Sensitivity for Binary Treatments

7.1 The Limitation of Constant Effects

The partial linearity assumption $\mathbb{E}[Y \mid D, X, U] = \theta D + g(X, U)$ implies a **constant** treatment effect θ . While this is a useful simplification for deriving the theory, it is a strong assumption in many practical settings, especially for binary treatments.

Consider our hospital example. The treatment is binary: either a message is sent ($D = 1$) or it is not ($D = 0$). If the treatment effect is constant, then the message has the same effect on every patient, regardless of their characteristics. This would mean that we should either send the message to everyone or to no one. But the entire motivation for building the monitoring system was to *target* who to send a message to, which implicitly assumes that the effect varies across patients. So, the constant effect assumption is not reasonable in this example.

For most practical scenarios involving binary treatments, we need to allow for **heterogeneous treatment effects**. The question then arises: can we perform a similar sensitivity analysis without assuming a constant effect? The answer is yes. A framework exists for binary treatments that allows for arbitrary treatment effect heterogeneity.

7.2 The Doubly Robust Estimator Under Confounding

Since we are not making a partial linearity assumption, the appropriate estimator when controlling for a subset X of the control variables is the **doubly robust (DR) estimator**, not DML. Recall that the DR estimator allows for heterogeneous treatment effects and does not require the outcome to be linear in the treatment.

If we run the DR estimator on the subset X (omitting U), we will obtain an estimate that converges to some value θ_s , which is not the true average treatment effect (ATE). The question is: what is the bias of this estimate?

7.3 A New Confounding Measure for the Treatment

The bias can still be bounded by a product of two terms: one related to the confounding on the outcome and one related to the confounding on the treatment. The measure of confounding on the outcome, $\eta_{Y \sim U|D,X}^2$, remains exactly the same as before. However, the measure of confounding on the treatment changes in this more general, non-linear setting.

Definition: Gain in Precision for Treatment (General Setting)

$$\eta_{D \sim U|X}^2 = \frac{\mathbb{E}[1/\text{Var}(D | X, U)] - \mathbb{E}[1/\text{Var}(D | X)]}{\mathbb{E}[1/\text{Var}(D | X, U)]} \in [0, 1]$$

Interpretation: How much more **precise** our prediction of the treatment D becomes when we add the unobserved confounder U to our model. Here, precision is defined as the inverse of the variance, $1/\text{Var}(D | \cdot)$.

This is a subtle but important change. Instead of measuring the *reduction in variance* (as in the PLR setting), we now measure the *gain in precision* (the inverse of variance). The distinction matters because of Jensen's inequality: $\mathbb{E}[1/\text{Var}(D | X, U)]$ is not the same as $1/\mathbb{E}[\text{Var}(D | X, U)]$.

7.4 Bias Formula for Binary Treatments

With this new measure, the bias formula is analogous to the one we derived before:

Bias Formula for Binary Treatments

If we use the Doubly Robust estimator (which allows for heterogeneous treatment effects) instead of DML, we converge to a biased estimate θ_s such that:

$$|\theta_s - \theta|^2 \leq \eta_{Y \sim U|D,X}^2 \cdot \frac{\eta_{D \sim U|X}^2}{1 - \eta_{D \sim U|X}^2} \cdot S^2$$

where the scaling factor S^2 is now:

$$S^2 = \mathbb{E}[\text{Var}(Y | D, X)] \cdot \mathbb{E}[1/\text{Var}(D | X)]$$

and can be estimated from data.

The structure is the same: the bias is bounded by a product of the confounding strength on the outcome, the confounding strength on the treatment, and a data-estimable scaling factor. The only changes are in how we measure the confounding on the treatment (gain in precision instead of reduction in variance) and in the form of the scaling factor. The same style of analysis applies: we run the DR estimator on the observed controls, posit bounds on the sensitivity parameters, estimate S^2 from the data, and compute the final confidence intervals.

7.5 Applying to the Hospital Example

In the hospital setting, we can now reason about the bounds without any strong assumptions about constant effects. We need to quantify two things:

1. **Bound on η_Y^2 :** How much variance in patient outcomes (time in range) could physician workload explain, on top of patient data and whether they got a message? Given that the outcome is primarily driven by the patient's health state and the message content, the physician's workload likely explains only a small additional fraction. *Probably small.*
2. **Bound on η_D^2 :** How much more precise would our prediction of "message sent" be if we knew the physician's workload? The workload would be a factor in the decision, but the decision is primarily driven by patient monitoring data. *It would be a factor, but likely not a substantial one.*

With these qualitative assessments translated into quantitative bounds, the product formula gives us bounds on the treatment effect of sending a message.

7.6 Comparing the Two Confounding Measures

It is instructive to compare the confounding measure for the treatment in the general setting with the one from the partially linear setting:

PLR Setting	General Setting
$\eta_{D \sim U X}^2 = \frac{\mathbb{E}[\text{Var}(D X)] - \mathbb{E}[\text{Var}(D X, U)]}{\mathbb{E}[\text{Var}(D X)]}$	$\eta_{D \sim U X}^2 = \frac{\mathbb{E}[1/\text{Var}(D X, U)] - \mathbb{E}[1/\text{Var}(D X)]}{\mathbb{E}[1/\text{Var}(D X, U)]}$
Measures reduction in variance	Measures gain in precision

If we were to use the naive (incorrect) approximation $\mathbb{E}[1/\text{Var}(D | Z)] \approx 1/\mathbb{E}[\text{Var}(D | Z)]$ for each of the terms in the general setting, we would recover the measure from the PLR setting. The reason the general setting uses precision rather than variance is deeply connected to the flexibility of the outcome model.

The Price of Flexibility

The term $\mathbb{E}[1/\text{Var}(D | Z)]$ is more sensitive to the addition of elements in the conditioning set Z than the term $1/\mathbb{E}[\text{Var}(D | Z)]$. In the general setting, the unobserved confounder can cause large omitted variable bias by simply reducing the variance of the treatment for a small region of X 's. This will cause the denominator $\text{Var}(D | X, U)$ to be very small for these X 's, leading to large changes in precision gained. This is the **price we have to pay for allowing a very flexible outcome regression model**: confounders that explain the treatment in some small region of X 's can cause large omitted variable bias, even if their overall predictive power is modest.

This comparison highlights an important trade-off. The partially linear model is more restrictive (it assumes constant effects) but produces tighter sensitivity bounds. The general binary treatment framework is more flexible (it allows heterogeneous effects) but the sensitivity bounds may be wider because the precision-based measure is more sensitive to localized confounding.

In the paper linked at the end of this chapter (Chernozhukov, Cinelli, Newey, Sharma, and Syrgkanis, 2021), bias bounds have been produced for many other causal quantities of interest. All of these bias bounds typically have the same product structure: one factor related to how the confounder predicts the outcome, and another related to how the confounder predicts the treatment. The two examples we have discussed—the partially linear model and the binary treatment model—are the leading applications of this framework.

8 Part VI: Beyond Conditioning — The Front-Door Criterion

All of the sensitivity analysis we have developed so far has been a byproduct of our singular focus on identifying treatment effects by conditioning on observables. So far in this course, that is the

only method we have used for identification. We perform sensitivity analysis precisely because conditioning on observables is the only tool in our toolkit, and when it fails, we need to quantify how badly it might fail.

However, when approaching a new empirical problem, the first step should always be to ask for more information. But the next step should *not* be to jump immediately to sensitivity analysis. Instead, we should ask: is there a more clever way to identify the treatment effect beyond simply conditioning on observables? This question leads us to a broader view of what identification means.

8.1 Back to Basics: What is Identification?

Let us revisit the fundamental question: what is identification of a causal effect?

Definition: Identification

Identification is the exercise of expressing a causal quantity of interest (e.g., the ATE) as a **statistical quantity**—a function of the joint probability distribution of the observed variables—that can be measured from observational data.

We start with a formula for the average treatment effect (ATE), which depends on interventional or potential outcomes:

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)]$$

We do not know this quantity because it is based on outcomes that are not observed. The goal of identification is to express this unknown quantity as some function of the things we *do* observe. What we observe is the joint probability distribution of all our measured variables, say $V_1, \dots, V_k$. In the infinite sample limit, we can think of this distribution as being known to us. So the question becomes: once we know this distribution, can we write the ATE as some function of this distribution? If we can, then we can estimate the distribution from our finite sample, plug it into the formula, and we are done.

So far, the only identification formulas we have seen are the **G-formula** and the **IPW formula**. These are formulas that depend only on the probability distribution of observable random variables. For example:

$$\text{ATE} = \mathbb{E}_X[\mathbb{E}[Y \mid D = 1, X] - \mathbb{E}[Y \mid D = 0, X]]$$

The conditional expectations and the averaging are all functions of the observable distribution. However, the problem of identification is not just about finding a set of covariates X for which we can write such a formula. That is a very narrow perspective.

The broader question is: given our assumptions about how the world works (expressed as a DAG with observed and unobserved variables, or as assumptions about functional forms), does there exist *any* formula that allows us to write the unknown causal quantity as a function of the probability distribution of the observed random variables? This is a much broader question, and in fact, there are many identification formulas that do not rely on conditioning on observables. The

front-door criterion is one such formula.

8.2 The Front-Door Criterion: An Example from Ride-Sharing

To illustrate the front-door criterion, we consider a recent and compelling empirical example from Bellemare, Bloem, and Wexler (2024) on ride-sharing platforms.

The Uber/Lyft Tipping Example

The research question is: does a passenger **authorizing** a shared ride (e.g., choosing UberX Share) have any effect on the **tips** that the passenger gives at the end of the ride?

The key variables are:

- D : Passenger **authorizes** a shared ride.
- M : Passenger **actually shares** the ride with another person.
- X : Time of day, source region, and target region.
- Y : Tipping behavior (e.g., tip percentage).

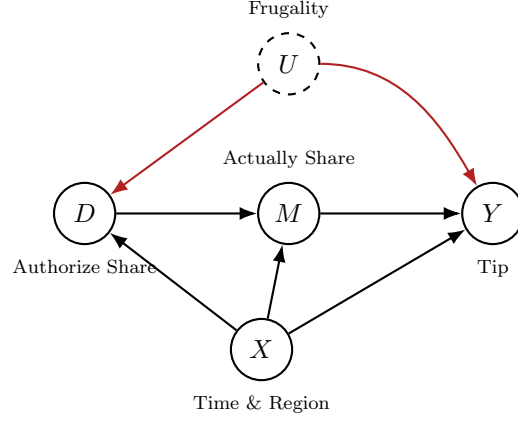
The problem: Frugal people are more likely to authorize sharing ($U \rightarrow D$) and also more likely to tip less ($U \rightarrow Y$). We have unobserved confounding from the passenger’s frugality U .

The key insight: Authorizing a shared ride should not have any direct effect on tips unless a shared ride *actually occurs*. The fact that a passenger says “I’m okay with a shared ride” should not affect their tipping behavior unless Uber’s algorithm actually assigns them a shared ride. The variable M (actually sharing) fully **mediates** the effect of D on Y .

Moreover, whether a shared ride actually occurs depends on Uber’s algorithm, which uses characteristics like time of day, source region, and target region (X). Crucially, the passenger’s frugality (U) does *not* enter Uber’s algorithm—the algorithm does not know or care about the passenger’s financial disposition. Therefore, U has no direct effect on M (conditional on D and X).

8.3 The Front-Door DAG

This scenario corresponds to a specific causal graph structure:



The confounding path $D \leftarrow U \rightarrow Y$ is open.
We cannot use the adjustment formula.

The red arrows highlight the confounding: the unobserved frugality U creates a backdoor path $D \leftarrow U \rightarrow Y$ that cannot be blocked by conditioning on X alone. Despite this, we will show that the causal effect of D on Y is identifiable.

8.4 The Three Front-Door Conditions

For the front-door criterion to apply, we need a special structure involving a mediating variable M :

The Front-Door Conditions

The front-door criterion holds if there exists a variable (or set of variables) M such that:

1. **Full mediation:** M intercepts **all** directed paths from D to Y . That is, the only way D can affect Y is through M (i.e., $D \rightarrow M \rightarrow Y$).
2. **No unblocked backdoor from D to M :** There is no unblocked backdoor path from D to M , after conditioning on X . That is, D is as-if random with respect to M , conditional on X .
3. **Backdoor from M to Y blocked by $\{D, X\}$:** All backdoor paths from M to Y are blocked by conditioning on $\{D, X\}$.

In our DAG, the variable M (Actually Share) satisfies all three conditions. Let us verify each:

1. **Full mediation:** The only directed path from D to Y is $D \rightarrow M \rightarrow Y$. There is no direct edge $D \rightarrow Y$, which encodes the assumption that authorizing a shared ride has no effect on tipping except through actually sharing.
2. **No unblocked backdoor from D to M :** The potential backdoor paths from D to M would need to go through U . But U has no edge to M (Uber's algorithm does not use the passenger's frugality), so there is no backdoor path from D to M that involves U . Conditioning on X blocks any remaining paths through the observed covariates.

3. **Backdoor from M to Y blocked by $\{D, X\}$:** The backdoor path from M to Y goes through $M \leftarrow D \leftarrow U \rightarrow Y$. Conditioning on D blocks this path at the chain $M \leftarrow D$.

8.5 How the Front-Door Criterion Works

The front-door criterion allows us to identify the total effect of D on Y by decomposing it into two parts that *are* separately identified.

8.5.1 Step 1: Identify the Effect of D on M

Step 1: The Effect of D on M

$$\gamma = \text{Effect}(D \rightarrow M) = \mathbb{E}[\mathbb{E}[M \mid D = 1, X] - \mathbb{E}[M \mid D = 0, X]]$$

Why is this identified? Because the relationship between D and M is **unconfounded** conditional on X . There is no open backdoor path from D to M once we condition on X . We can simply regress M on D and X .

This step uses the standard identification-by-conditioning approach that we have been using throughout the course. The key insight is that even though D is confounded with Y , it is *not* confounded with M (conditional on X). The confounding from U does not affect the $D \rightarrow M$ relationship because U has no edge to M .

8.5.2 Step 2: Identify the Effect of M on Y

Step 2: The Effect of M on Y

$$\delta = \text{Effect}(M \rightarrow Y) = \mathbb{E}[\mathbb{E}[Y \mid M = 1, D, X] - \mathbb{E}[Y \mid M = 0, D, X]]$$

Why is this identified? Because all backdoor paths from M to Y (e.g., $M \leftarrow D \leftarrow U \rightarrow Y$) are **blocked by conditioning on $\{D, X\}$** .

This step is more subtle. The backdoor path from M to Y goes through D and U . But by conditioning on D , we block this path. This is a bit peculiar: we are using the treatment D as a *control variable* in this second regression. We are looking at the population of people who authorized the share, were from a specific source and target region, and at a certain time of day. For this population, we analyze what happens when we flip the variable M from on to off.

8.5.3 Step 3: Combine the Two Effects

The total causal effect of D on Y is fully mediated through M . Therefore, we can express the total effect as the product of the two identified effects:

The Front-Door Formula (Linear Case)

$$\theta = \gamma \cdot \delta$$

Operationally, for linear models:

1. Run OLS($M \sim D, X$); let $\hat{\gamma}$ be the coefficient on D .
2. Run OLS($Y \sim M, D, X$); let $\hat{\delta}$ be the coefficient on M .
3. Return $\hat{\theta} = \hat{\gamma} \cdot \hat{\delta}$.

If we do not want to assume linear models, we should use more robust methods. For instance, we could use a doubly robust estimator for the first stage (D 's effect on M) and another doubly robust estimator for the second stage (M 's effect on Y), and then take the product of those two estimates.

A Subtle Assumption: Independence of Individual Effects

The simple product-of-averages formula $\theta = \gamma \cdot \delta$ requires an extra subtle assumption: the individual-level effect of D on M must be independent of the individual-level effect of M on Y . If these two effects are not independent when viewed as random variables across the population, then the average of the product of the effects will not equal the product of their averages. Under linear specifications, the causal effects are constant across the population, making them trivially independent. Therefore, the two-stage OLS algorithm is valid under a constant treatment effect assumption. However, if we are not willing to make such a strong assumption, we should not use this simplistic calculation. It is worth noting that the empirical paper by Bellemare, Bloem, and Wexler (2024) did exactly this: they ran two OLS regressions and took the product of the coefficients.

8.6 The General Front-Door Formula

More generally, when we are not willing to assume linearity, we can estimate the effect of D on the entire *distribution* of M , and the effect of M on the entire *distribution* of Y , and then combine them. The general front-door formula expresses the interventional distribution of Y under an intervention on D :

The General Front-Door Formula

$$P(Y(d) = y) = \sum_x p(x) \sum_m P(M = m \mid D = d, X = x) \sum_{d'} P(Y = y \mid M = m, D = d', X = x) P(D = d' \mid X = x)$$

The intuition behind this formula is that the causal effect is a weighted combination of the effect of $D \rightarrow M$ and the effect of $M \rightarrow Y$. The first factor, $P(M = m \mid D = d, X = x)$, captures how the treatment shifts the distribution of the mediator. The second factor, $\sum_{d'} P(Y = y \mid M = m, D = d', X = x)P(D = d' \mid X = x)$, captures the effect of the mediator on the outcome, averaged over the marginal distribution of the treatment. The simplistic idea of just multiplying the average effects only works in linear models without interaction.

Pattern for Front-Door Identification

More generally, whenever you have a **total mediator** that is **unconfounded** given conditional observables, you can use the front-door identification strategy. In the ride-sharing example, the mediator was the algorithmic decision of whether or not to implement the shared ride that the user authorized. Whenever a user action requires authorization from a system's algorithm to have a final effect, this specific structure arises. The system's algorithm does not account for the unobserved confounding variables that influence the user's initial decision, creating exactly the conditions needed for front-door identification.

8.7 The Broader Point: Many Identification Strategies Exist

The front-door criterion is just one example of an identification strategy that does not rely on observing all confounders. In fact, given any causal DAG, there is a complete algorithm that can determine if a causal effect is identifiable.

The ID Algorithm

The **ID algorithm** (e.g., implemented in packages like `DoWhy` in Python) is a black-box algorithm that takes as input a causal DAG and automatically determines whether a given causal effect is identifiable from the observed data. If the effect is identifiable, the algorithm returns the correct statistical formula to compute it. If not, it reports that the effect is not identifiable from the graphical assumptions alone.

It is important to understand that the ID algorithm exclusively accepts **graphical assumptions** as input. The only information it utilizes pertains to the parental relationships defined within the graph structure. If we are willing to introduce additional assumptions—such as specifying the functional forms of the structural equations (e.g., linearity)—the algorithm might incorrectly determine that a quantity is not identifiable when, in fact, it is under those extra assumptions.

8.8 What if the Effect is Not Identifiable?

What happens if the ID algorithm fails and tells us the effect is not identifiable from the graph alone? This means we cannot identify the effect without making **further assumptions** beyond the parental relationships in the DAG. We have several options:

1. **Impose extra restrictions on the structural equations.** For example, we might assume linearity, monotonicity, or other functional form restrictions. These go beyond the information in the causal graph but may be justified by domain knowledge.
2. **Re-interpret the causal quantity.** Perhaps the exact quantity we are after is not identifiable, but a closely related quantity is. For example, we might not be able to identify the ATE, but we might be able to identify the average effect for some sub-group of the population.
3. **Use sensitivity analysis.** If identification fails entirely, we can fall back on the sensitivity analysis framework developed earlier in this lecture to bound the effect.

Many techniques in econometrics rely on such auxiliary assumptions. In the next lecture, we will explore **instrumental variables**, which is one of the most powerful and widely used identification strategies for dealing with unobserved confounding. Instrumental variables impose specific restrictions on the structural equations (specifically, the existence of a variable that affects the treatment but has no direct effect on the outcome) and, in some formulations, redefine the causal quantity of interest to achieve identification.

9 Summary and Key Takeaways

This lecture addressed the fundamental challenge of unobserved confounding—the situation where conditional exogeneity fails because we cannot observe or control for all relevant variables. We developed two complementary strategies for dealing with this challenge.

1. **Sensitivity analysis bounds the effect.** When unobserved confounding is present, we cannot obtain a point estimate of the causal effect. However, we can provide **bounds** on the true effect by making explicit assumptions about the confounding strength. These bounds reflect the fundamental uncertainty about the unobserved confounder, not just statistical noise.
2. **Bias is characterized by intuitive quantities.** The omitted variable bias is characterized by gains in explained variance—the partial R^2 quantities η_Y^2 and η_D^2 . These provide an intuitive way for domain experts to reason about confounding: “How much more of the outcome/treatment variance could an unobserved confounder explain?” The bias is always a product of these two quantities, meaning that a confounder must be strongly predictive of *both* the treatment and the outcome to cause large bias.
3. **Benchmarking provides scale.** To make sensitivity analysis practical, we can benchmark the required confounding strength against the strength of observed covariates. Contour plots visualize the robustness of our findings, and the robustness value provides a single, interpretable summary metric.

4. **Identification is more than just adjustment.** The front-door criterion demonstrates that other causal structures allow for identification even with unobserved confounding. The key insight is that if the treatment’s effect is fully mediated through a variable that is itself unconfounded, we can decompose the total effect into two identified pieces. More broadly, the ID algorithm can automatically determine whether a causal effect is identifiable from any given DAG. Always consider the full causal graph to see if alternative identification strategies apply before resorting to sensitivity analysis.

In the next lecture, we will introduce **instrumental variables**—a powerful and widely used identification strategy for unobserved confounding that relies on finding a special variable (the instrument) with specific properties. This will further expand our toolkit beyond conditioning and the front-door criterion.

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A Standard Errors for Sensitivity Bounds via the Delta Method

The confidence interval for the sensitivity bounds relies on being able to compute the standard errors for the upper and lower bound estimates, $\hat{\theta}_{\text{upper}}$ and $\hat{\theta}_{\text{lower}}$. Since these bounds are functions of other asymptotically linear estimators (namely $\hat{\theta}_s$, $\hat{\sigma}^2$, and $\hat{\nu}^2$), we can use the **delta method** to derive their influence functions and, consequently, their standard errors.

Recall that an estimator $\hat{\alpha}$ is asymptotically linear if it can be written as $\hat{\alpha} - \alpha = \frac{1}{n} \sum_{i=1}^n \psi_{\alpha}(Z_i) + o_p(1/\sqrt{n})$, where ψ_{α} is the **influence function**. The influence functions for our three base estima-

tors are:

$$\begin{aligned}\psi_{\theta_s}(Z_i) &= \frac{1}{\nu^2} \tilde{D}_i (\tilde{Y}_i - \theta_s \tilde{D}_i) \\ \psi_{\sigma^2}(Z_i) &= (\tilde{Y}_i - \theta_s \tilde{D}_i)^2 - \sigma^2 \\ \psi_{\nu^2}(Z_i) &= \tilde{D}_i^2 - \nu^2\end{aligned}$$

The upper bound is given by the function $g(\theta_s, \sigma^2, \nu^2) = \theta_s + \sqrt{\frac{\sigma^2}{\nu^2}} \cdot C$, where $C = \sqrt{\bar{\eta}_Y^2 \frac{\bar{\eta}_D^2}{1 - \bar{\eta}_D^2}}$ is a constant determined by the sensitivity parameters. The delta method states that the influence function for g is the sum of the partial derivatives of g with respect to each of its arguments, multiplied by the influence function of each argument:

$$\psi_g(Z_i) = \frac{\partial g}{\partial \theta_s} \psi_{\theta_s}(Z_i) + \frac{\partial g}{\partial \sigma^2} \psi_{\sigma^2}(Z_i) + \frac{\partial g}{\partial \nu^2} \psi_{\nu^2}(Z_i)$$

Let's compute the partial derivatives:

$$\begin{aligned}\frac{\partial g}{\partial \theta_s} &= 1 \\ \frac{\partial g}{\partial \sigma^2} &= \frac{C}{2\sqrt{\sigma^2 \nu^2}} = \frac{C}{2S\nu^2} \\ \frac{\partial g}{\partial \nu^2} &= -\frac{C\sqrt{\sigma^2}}{2(\nu^2)^{3/2}} = -\frac{CS}{2\nu^2}\end{aligned}$$

Plugging these into the delta method formula gives us the influence function for the upper bound, ψ_{upper} . The influence function for the lower bound, ψ_{lower} , is analogous, with the sign of the second and third terms flipped. The standard error for each bound is then simply the standard deviation of its influence function, scaled by $\sqrt{n}$:

$$\text{SE}_{\text{upper}} = \sqrt{\frac{\text{Var}(\psi_{\text{upper}}(Z))}{n}}$$

This is precisely what is implemented in the Python code in Section 5.3 to compute the standard errors for the sensitivity interval.